```
                postgresql://username:password@hostname/dbname
                or an existing connection: dict_keys(['mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Serv
er&trusted_connection=yes'])
```

In [33]:
```
%config SqlMagic.style = '_DEPRECATED_DEFAULT'
```

EDA

1- top 5 row from data

2- total of booking

3- confirmed booking

4- cancelled booking

5- total booking nights

6- avrage of staying at hotel

7- percent of Repeat Customer

8- Average of Spacial Request

9- Average Advance booking period in days

10- Overall cancellation rate by hotel, month, and market segment

11- Cancellation rate by channel and agent

12- Cancellation rate for customers who have had previous cancellations

13- Cancellation rate by deposit type

14- Average number of days between booking and arrival for both cancelled and uncancelled bookings

15- Monthly cancellation rate

1- Top 5 Row From Data

In [61]:
```
%%sql
select top(5)* from data2018;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[61]:

hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_week_number	arrival_date_day_of_month	stays_in_weeken
Resort Hotel	1	84	2018	August	31	1	
Resort Hotel	1	84	2018	August	31	1	
Resort Hotel	1	89	2018	August	31	1	
Resort Hotel	1	10	2018	August	31	1	
Resort Hotel	1	80	2018	August	31	1	

2- Total Of Booking

In [36]:
```
%%sql
select distinct hotel,count(reservation_status_date) as total_booking
```

```
from data2018 group by hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[36]:

hotel	total_booking
City Hotel	13682
Resort Hotel	8314

3- Total Confirmed Booking

In [62]:
```
%%sql
select distinct hotel,count(reservation_status_date) as total_booking_confirmed
from data2018 where is_canceled = 0 group by hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[62]:

hotel	total_booking_confirmed
City Hotel	7678
Resort Hotel	6176

4- Total Cancelled Booking

In [63]:
```
%%sql
select distinct hotel,count(reservation_status_date) as total_booking_cancelled
from data2018 where is_canceled = 1 group by hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[63]:

hotel	total_booking_cancelled
City Hotel	6004
Resort Hotel	2138

5- Total Booking Nights

In [64]:
```
%%sql
select hotel,count(stays_in_weekend_nights)+count(stays_in_week_nights) as booking_nights
from data2018 group by hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[64]:

hotel	booking_nights
City Hotel	27364
Resort Hotel	16628

6- Avrage Days Of Staying At Hotel

In [42]:
```
%%sql
select hotel,(count(stays_in_weekend_nights)+count(stays_in_week_nights))/count(reservation_status_date) as avg_stays
from data2018 group by hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[42]:

hotel	avg_stays
City Hotel	2
Resort Hotel	2

7- Percentage Of Repeat Customer

In [65]:
```
%%sql
select hotel,cast(round(COUNT(CASE WHEN is_repeated_guest = 1 THEN 1 END) * 1.0
    / NULLIF(COUNT(reservation_status_date),0),2) as decimal(11,2)) as Loyalty_Customer
FROM data2018 group by hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[65]:

hotel	Loyalty_Customer
City Hotel	0.04
Resort Hotel	0.02

8- Average Of Spacial Request

In [55]:
```
%%sql
select hotel,sum(total_of_special_requests) as special_requests FROM data2018
group by hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[55]:

hotel	special_requests
City Hotel	4458
Resort Hotel	5120

9- Average Advance Booking Period In Days

In [67]:
```
%%sql
select hotel, avg(lead_time) as special_requests FROM data2018
group by hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[67]:

hotel	special_requests
City Hotel	105
Resort Hotel	83

10- Overall Cancellation Rate By Hotel, Month, And Market Segment

In [46]:
```
%%sql
SELECT
    hotel,
    arrival_date_month,
    market_segment,
    cast(round(COUNT(CASE WHEN is_canceled = 1 THEN 1 END) *1.0
        / NULLIF(COUNT(reservation_status_date), 0),2)as decimal(12,2)) as Avg_Cancelation
FROM data2018
GROUP BY
    hotel,
    arrival_date_month,
    market_segment
ORDER BY
    hotel,
    arrival_date_month,
    market_segment;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[46]:

hotel	arrival_date_month	market_segment	Avg_Cancelation
City Hotel	August	Complementary	0.03
City Hotel	August	Corporate	0.29
City Hotel	August	Direct	0.10
City Hotel	August	Groups	0.73
City Hotel	August	Offline TA/TO	0.56
City Hotel	August	Online TA	0.20
City Hotel	August	Undefined	1.00
City Hotel	December	Complementary	0.03
City Hotel	December	Corporate	0.11
City Hotel	December	Direct	0.23
City Hotel	December	Groups	0.62
City Hotel	December	Offline TA/TO	0.54
City Hotel	December	Online TA	0.31
City Hotel	July	Complementary	0.20
City Hotel	July	Corporate	0.00
City Hotel	July	Direct	0.48
City Hotel	July	Groups	0.74
City Hotel	July	Offline TA/TO	0.36
City Hotel	July	Online TA	0.82
City Hotel	November	Complementary	0.22
City Hotel	November	Corporate	0.24
City Hotel	November	Direct	0.25

hotel	arrival_date_month	market_segment	Avg_Cancelation
City Hotel	November	Groups	0.31
City Hotel	November	Offline TA/TO	0.14
City Hotel	November	Online TA	0.25
City Hotel	October	Complementary	0.33
City Hotel	October	Corporate	0.05
City Hotel	October	Direct	0.22
City Hotel	October	Groups	0.63
City Hotel	October	Offline TA/TO	0.38
City Hotel	October	Online TA	0.19
City Hotel	September	Complementary	0.00
City Hotel	September	Corporate	0.30
City Hotel	September	Direct	0.10
City Hotel	September	Groups	0.66
City Hotel	September	Offline TA/TO	0.43
City Hotel	September	Online TA	0.17
Resort Hotel	August	Corporate	0.44
Resort Hotel	August	Direct	0.19
Resort Hotel	August	Groups	0.06
Resort Hotel	August	Offline TA/TO	0.18
Resort Hotel	August	Online TA	0.33
Resort Hotel	December	Complementary	0.14
Resort Hotel	December	Corporate	0.01

hotel	arrival_date_month	market_segment	Avg_Cancelation
Resort Hotel	December	Direct	0.11
Resort Hotel	December	Groups	0.42
Resort Hotel	December	Offline TA/TO	0.14
Resort Hotel	December	Online TA	0.30
Resort Hotel	July	Complementary	0.00
Resort Hotel	July	Corporate	0.17
Resort Hotel	July	Direct	0.12
Resort Hotel	July	Groups	0.58
Resort Hotel	July	Offline TA/TO	0.15
Resort Hotel	July	Online TA	0.30
Resort Hotel	November	Complementary	0.00
Resort Hotel	November	Corporate	0.22
Resort Hotel	November	Direct	0.04
Resort Hotel	November	Groups	0.33
Resort Hotel	November	Offline TA/TO	0.08
Resort Hotel	November	Online TA	0.20
Resort Hotel	October	Complementary	0.00
Resort Hotel	October	Corporate	0.10
Resort Hotel	October	Direct	0.12
Resort Hotel	October	Groups	0.49
Resort Hotel	October	Offline TA/TO	0.24
Resort Hotel	October	Online TA	0.16

hotel	arrival_date_month	market_segment	Avg_Cancelation
Resort Hotel	September	Complementary	0.00
Resort Hotel	September	Corporate	0.13
Resort Hotel	September	Direct	0.19
Resort Hotel	September	Groups	0.67
Resort Hotel	September	Offline TA/TO	0.25
Resort Hotel	September	Online TA	0.33

11- Cancellation Rate By Channel And Agent

In [68]:
```
%%sql
SELECT
    agent,
    customer_type,
    cast(round(COUNT(CASE WHEN is_canceled = 1 THEN 1 END) *1.0
        / NULLIF(COUNT(reservation_status_date), 0),2)as decimal(12,2)) as Avg_Cancelation
FROM data2018
GROUP BY
    agent,
  customer_type
ORDER BY
    agent,
    customer_type;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[68]:

agent	customer_type	Avg_Cancelation
1	Contract	1.00
1	Transient	0.97
1	Transient-Party	0.54
10	Group	0.00
10	Transient	0.34
10	Transient-Party	0.00
103	Transient	0.00
104	Transient	0.05
104	Transient-Party	0.00
105	Transient	0.10
105	Transient-Party	1.00
106	Group	0.00
11	Group	0.00
11	Transient	0.14
110	Transient	0.50
111	Transient-Party	0.00
112	Transient-Party	0.00
114	Transient	0.00
115	Contract	0.00
115	Group	0.00
115	Transient	0.06
115	Transient-Party	0.00

agent	customer_type	Avg_Cancelation
117	Transient	0.00
118	Transient	0.00
119	Transient-Party	0.18
12	Transient	1.00
12	Transient-Party	0.13
121	Transient	0.00
126	Transient	0.00
127	Transient	0.00
128	Contract	0.00
13	Contract	0.09
13	Group	1.00
13	Transient	0.33
13	Transient-Party	0.38
134	Transient	0.54
134	Transient-Party	0.71
135	Transient	0.00
138	Transient	0.00
14	Contract	1.00
14	Group	0.08
14	Transient	0.17
14	Transient-Party	0.23
142	Contract	0.22

agent	customer_type	Avg_Cancelation
142	Group	0.00
142	Transient	0.11
142	Transient-Party	1.00
143	Contract	0.25
143	Transient	0.00
146	Transient	0.27
146	Transient-Party	0.00
147	Transient	0.06
147	Transient-Party	0.00
148	Transient	0.00
149	Transient	0.31
149	Transient-Party	1.00
15	Group	0.00
15	Transient	0.38
15	Transient-Party	0.17
151	Transient-Party	0.00
152	Transient	0.17
152	Transient-Party	0.00
155	Transient-Party	0.48
156	Contract	0.07
156	Transient	0.10
156	Transient-Party	0.00

agent	customer_type	Avg_Cancelation
158	Transient	0.00
16	Transient	0.16
167	Transient	0.00
168	Transient-Party	0.00
17	Transient	0.00
17	Transient-Party	0.49
171	Transient	0.13
171	Transient-Party	0.00
175	Group	0.00
175	Transient	0.13
175	Transient-Party	0.29
177	Transient	0.06
177	Transient-Party	0.29
181	Transient	0.20
181	Transient-Party	0.00
182	Transient	0.00
183	Transient	0.94
183	Transient-Party	0.00
184	Transient	0.00
185	Transient-Party	0.12
19	Transient	0.99
19	Transient-Party	0.09

agent	customer_type	Avg_Cancelation
193	Transient	0.00
195	Transient	1.00
196	Transient	0.22
196	Transient-Party	0.25
2	Contract	0.04
2	Transient	0.17
2	Transient-Party	0.43
20	Transient	0.89
20	Transient-Party	0.16
201	Transient-Party	0.10
208	Transient	0.96
208	Transient-Party	0.18
21	Transient	0.95
21	Transient-Party	0.24
22	Transient	0.08
22	Transient-Party	0.00
24	Transient-Party	0.00
240	Group	0.07
240	Transient	0.31
240	Transient-Party	0.20
241	Group	0.00
241	Transient	0.12

agent	customer_type	Avg_Cancelation
242	Transient	0.30
242	Transient-Party	1.00
243	Contract	0.03
243	Transient	0.03
243	Transient-Party	0.00
244	Transient	0.00
245	Transient-Party	0.08
248	Transient	1.00
248	Transient-Party	0.11
250	Contract	0.08
250	Group	0.00
250	Transient	0.16
250	Transient-Party	0.07
251	Transient	0.11
251	Transient-Party	0.00
252	Transient	1.00
253	Transient	0.00
256	Transient	0.00
256	Transient-Party	0.09
258	Transient	0.00
26	Contract	0.07
26	Transient	0.08

agent	customer_type	Avg_Cancelation
26	Transient-Party	0.02
261	Transient	0.24
27	Group	0.00
27	Transient	0.07
273	Group	0.00
273	Transient	1.00
273	Transient-Party	0.11
275	Transient	0.00
275	Transient-Party	0.00
28	Contract	0.08
28	Group	0.00
28	Transient	0.05
28	Transient-Party	0.00
281	Contract	0.36
281	Transient	1.00
281	Transient-Party	1.00
29	Transient	0.98
29	Transient-Party	0.11
3	Transient	0.93
3	Transient-Party	0.19
30	Transient	1.00
30	Transient-Party	0.27

agent	customer_type	Avg_Cancelation
300	Transient	0.00
301	Transient	0.00
303	Transient	0.00
304	Transient	0.00
305	Transient	0.00
305	Transient-Party	0.07
306	Transient	0.00
308	Transient	0.00
308	Transient-Party	0.04
31	Transient-Party	0.00
313	Transient	1.00
313	Transient-Party	0.03
314	Contract	0.04
314	Group	0.00
314	Transient	0.04
314	Transient-Party	0.35
32	Contract	1.00
32	Transient-Party	0.00
321	Transient	0.00
324	Transient	0.00
324	Transient-Party	0.14
326	Transient	0.00

agent	customer_type	Avg_Cancelation
326	Transient-Party	0.33
328	Transient	0.00
328	Transient-Party	0.00
330	Transient	0.00
330	Transient-Party	0.00
332	Transient	0.00
334	Group	0.00
334	Transient-Party	0.04
335	Transient-Party	0.00
336	Transient	0.00
339	Group	1.00
339	Transient	0.13
34	Transient	0.00
348	Transient	0.00
348	Transient-Party	0.00
350	Transient-Party	0.00
352	Transient	0.00
36	Transient	0.03
36	Transient-Party	0.00
37	Transient	0.90
37	Transient-Party	0.18
38	Transient	1.00

agent	customer_type	Avg_Cancelation
38	Transient-Party	0.52
39	Transient	0.00
39	Transient-Party	0.00
4	Transient-Party	0.59
40	Contract	0.07
40	Transient	0.02
40	Transient-Party	0.00
41	Transient	1.00
42	Transient	0.50
42	Transient-Party	0.00
44	Transient	1.00
44	Transient-Party	0.00
45	Group	0.00
45	Transient	0.20
47	Transient	0.50
5	Transient	0.58
5	Transient-Party	0.00
50	Transient	0.67
50	Transient-Party	0.00
52	Transient-Party	0.00
54	Transient	0.00
55	Transient	1.00

agent	customer_type	Avg_Cancelation
56	Transient-Party	0.06
57	Transient	0.00
6	Group	0.00
6	Transient	0.25
6	Transient-Party	0.59
60	Transient	1.00
61	Transient-Party	0.00
63	Transient	0.00
63	Transient-Party	0.00
64	Transient	1.00
66	Transient	0.00
66	Transient-Party	0.40
67	Transient	0.00
68	Transient	0.82
68	Transient-Party	0.14
69	Contract	0.00
69	Transient	0.00
7	Group	0.09
7	Transient	0.13
7	Transient-Party	0.08
70	Transient	0.00
71	Contract	0.00

agent	customer_type	Avg_Cancelation
71	Group	1.00
71	Transient	0.00
72	Transient	0.00
74	Transient	0.00
74	Transient-Party	0.00
75	Group	1.00
75	Transient	0.00
78	Transient	0.97
79	Transient	0.50
8	Contract	0.18
8	Transient	0.32
8	Transient-Party	0.27
81	Transient	0.17
82	Transient	1.00
82	Transient-Party	0.10
83	Contract	0.19
83	Group	0.00
83	Transient	0.09
83	Transient-Party	0.14
85	Group	0.00
85	Transient	0.08
85	Transient-Party	0.00

agent	customer_type	Avg_Cancelation
86	Transient	0.00
87	Group	1.00
87	Transient	0.33
87	Transient-Party	0.00
88	Transient	0.00
89	Transient	0.00
9	Contract	0.23
9	Group	1.00
9	Transient	0.32
9	Transient-Party	0.40
90	Transient	0.00
91	Transient	0.00
92	Transient-Party	0.00
94	Transient	0.00
95	Transient	0.20
95	Transient-Party	0.00
96	Group	1.00
96	Transient	0.79
96	Transient-Party	0.76
98	Transient	0.00
98	Transient-Party	0.31
99	Transient	0.00

agent	customer_type	Avg_Cancelation
99	Transient-Party	0.39
NULL	Contract	0.00
NULL	Group	0.60
NULL	Transient	0.24
NULL	Transient-Party	0.13

12- Cancellation Rate For Customers Who Have Had Previous Cancellations

In [71]:
```
%%sql
SELECT
    hotel,
    cast(round(COUNT(CASE WHEN is_canceled = 1 THEN 1 END) *1.0
        / NULLIF(COUNT(reservation_status_date), 0),2)as decimal(12,2)) as Avg_Cancelation
FROM data2018 where previous_cancellations =1
GROUP BY
  hotel
ORDER BY
    hotel;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[71]:

hotel	Avg_Cancelation
City Hotel	1.00
Resort Hotel	0.96

13- Cancellation Rate By Deposit Type

In [70]:
```
%%sql
SELECT
    deposit_type,
```

```
    cast(round(COUNT(CASE WHEN is_canceled = 1 THEN 1 END) *1.0
        / NULLIF(COUNT(reservation_status_date), 0),2)as decimal(12,2)) as Avg_Cancelation
FROM data2018
GROUP BY
  deposit_type
ORDER BY
    deposit_type;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[70]:

deposit_type	Avg_Cancelation
No Deposit	0.24
Non Refund	1.00
Refundable	0.00

14- Average Number Of Days Between Booking And Arrival For Cancelled and Uncancelled Bookings

In [51]:
```
%%sql
select is_canceled,avg(lead_time) as avg_lead_time_by_canceled_status
from data2018 group by is_canceled;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[51]:

is_canceled	avg_lead_time_by_canceled_status
0	60
1	160

15- Monthly Cancellation Rate

In [57]:
```
%%sql
select arrival_date_month,count(is_canceled) as cancelation_by_month
```

```
from data2018 where is_canceled = 1 group by arrival_date_month
order by cancelation_by_month desc;
```

```
 * mssql+pyodbc://@DESKTOP-73QP4NO/Hotels?driver=ODBC+Driver+17+for+sql+Server&trusted_connection=yes
Done.
```

Out[57]:

arrival_date_month	cancelation_by_month
September	2094
October	1732
August	1598
July	1259
December	973
November	486